MATH 151 B Homework 1-6

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Homework 1

1 a. Find the pointwise limit of $\frac{e^{x/n}}{n}$ for $x \in \mathbb{R}$

Fix $x \in \mathbb{R}$. We will claim that the function converges pointwise to 0. Next fix $\varepsilon > 0$. Choose $N > \frac{e^x}{\varepsilon}$. Thus we get that for all $n \geq N$ we get

$$\begin{aligned} |f_n(x) - 0| &= \left| \frac{e^{x/N}}{N} \right| \\ &= \frac{e^{x/N}}{N} \\ &\leq \frac{e^x}{N} \\ &< e^x \frac{\varepsilon}{e^x} \\ &= \varepsilon \end{aligned}$$

Thus the sequence converges pointwise $f_n(x) \to f(x) = 0$ on $\mathbb{R}$.

1 b. Is the limit uniform on $\mathbb{R}$?

By definition of uniform convergence. If $f_n \to f$ uniformly in $\mathbb{R}$, then for all $\varepsilon > 0$, There exists N such that for all $n \geq N$ we have

$$|f_n(x) - f(x)| < \varepsilon$$

For all $x \in \mathbb{R}$. Suppose that the function converges for some $x_1 \in \mathbb{R}$ Then we get that there exists N such that for all $n \geq N$

$$\left| \frac{e^{x/N}}{N} \right| < \varepsilon$$

Choose $\varepsilon = \frac{1}{2}$ and $x = n \ln n$. Thus we get for all n we get that

$$\begin{aligned} |f_n(x) - f(x)| &= \left| \frac{e^{\frac{n \ln n}{n}}}{n} - 0 \right| \\ &= \frac{n}{n} \\ &> \frac{1}{2} = \varepsilon \end{aligned}$$

Thus there exists some $\varepsilon > 0$ and $x \in \mathbb{R}$ such that $|f_n(x') - f(x')| > \varepsilon$. Meaning it's not uniformly convergent.

1 c. Is the limit uniform on $[0,1]$?

Fix $\varepsilon > 0$. We want to show that there exists N such that for all $n > N$ then

$$\sup_{x\in[0,1]} \left| \frac{e^{x/N}}{N} - 0 \right| < \varepsilon$$

The supremum occurs when $x = 1$ because the function is monotonically increasing so we need to show

$$\left| \frac{e^{1/N}}{N} - 0 \right| < \varepsilon$$

Choose $N > \frac{e}{\varepsilon}$. Thus we get

$$\begin{aligned} |f_n(1) - 0| &= \left| \frac{e^{1/N}}{N} \right| \\ &= \frac{e^{1/N}}{N} \\ &\leq \frac{e}{N} \\ &< e\frac{\varepsilon}{e} \\ &= \varepsilon \end{aligned}$$

Thus the sequence converges uniformly on $x \in [0,1]$

2. Find an example of a sequence of functions $\{f_n\}$ and $\{g_n\}$ that converge uniformly to some f and g on some set A, but such that $\{f_n g_n\}$ (the multiple) does not converge uniformly to fg on A

Let $f_n = g_n = x + \frac{1}{n}$. It converges uniformly to $f(x) = x$ in $\mathbb{R}$ because

$$\sup_{x\in\mathbb{R}} \left[x + \frac{1}{n} - x \right] = \frac{1}{n} \xrightarrow{n\to\infty} 0$$

Thus we get that the product

$$f_n g_n = \left(x + \frac{1}{n} \right)^2 = x^2 + 2\frac{x}{n} + \frac{1}{n^2} \qquad fg = x^2$$

Testing for uniform convergence

$$\sup_{x\in\mathbb{R}} \left[x^2 + 2\frac{x}{n} + \frac{1}{n^2} - x^2 \right] = \sup_{x\in\mathbb{R}} \left[2\frac{x}{n} + \frac{1}{n^2} \right]$$

If we choose $x = n$ we get that.

$$\left| 2\frac{n}{n} + \frac{1}{n^2} \right| > 2 > 0$$

We see that it's not uniformly convergent to x^2 on $\mathbb{R}$

3. Let $f_n : [0,1] \to \mathbb{R}$ be a sequence of increasing functions (that is $f_n(x) \geq f_n(y)$ whenever $x \geq y$). Suppose $f_n(0) = 0$ and $\lim_{n\to\infty} f_n(1) = 0$. Show that $\{f_n\}$ converges uniformly to 0

Since $f_n(x)$ is increasing and $f_n(0) = 0$ we see that for $x \in [0,1]$

$$0 = f_n(0) \leq f_n(x) \leq f_n(1)$$

Thus we see that the maximum of f_n on $[0,1]$ is at $x = 1$

$$\sup_{x\in[0,1]} f_n(x) = f_n(1)$$

$$\begin{aligned}\sup_{x\in[0,1]} |f_n - f| &= \sup_{x\in[0,1]} f_n(x) \\ &= f_n(1) \xrightarrow{n\to\infty} \\ &= 0\end{aligned}$$

Thus we see that $\{f_n\}$ converges uniformly to 0

5. Let $\{f_n\}$ be a sequence of functions defined on $[0,1]$. Suppose there exists a sequence of distinct numbers $x_n \in [0,1]$ such that

$$f_n(x_n) = 1$$

Prove or disprove the following statements:

True or False: There exists $\{f_n\}$ that converges to 0 pointwise

If we let $f_n = \frac{x}{n}$, then we see that f_n converges pointwise to 0 as $n \to \infty$. Next, let $x_n = n$. Composing f_n and x_n we get for all $n > 0$

$$f_n(x_n) = \frac{(n)}{n} = 1$$

Thus the fact is true

True or False: There exists $\{f_n\}$ that converges to 0 uniformly on $[0,1]$

Assume for contradiction that such a function exists. Then there exists N such that for all $n \geq N$ $|f_n(x) - 0| < \varepsilon$ for all $x \in [0,1]$. Choose $\varepsilon = \frac{1}{2}$ and let x_n be a sequence in $[0,1]$. By the requirements posed to us we get that

$$|f_n(x_n) - 0| = 1 < \varepsilon = \frac{1}{2}$$

A contradiction. Thus our assumption that the function existed was false.

Homework 2

1. While uniform convergence preserves continuity, it does not preserve differentiability. Find an explicit example of a sequence of differentiable functions on $[-1,1]$ that converge uniformly to a function f such that f is not differentiable.

Choose this wacky Fourier series

$$f_n(x) = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=1}^{n} \frac{\cos([2k-1]x)}{(2k-1)^2}$$

We get that $f_n(x)$ converges uniformly to $|x|$ on the interval $[-1,1]$ (it actually converges uniformly on a larger interval; $[-\pi,\pi]$). We know that $|x|$ is not differentiable at $x=0$.

2. Let $f:[0,1]\to\mathbb{R}$ be a Riemann integrable (hence bounded) function. Find

$$\lim_{n\to\infty}\int_0^1 \frac{f(x)}{n}dx$$

Define a constant $B := \sup_{x\in[0,1]} |f(x)|$ Thus we get that $f(x) \le B$ for all $x \in [0,1]$. Hence our integral becomes

$$\lim_{n\to\infty}\int_0^1 \frac{f(x)}{n}dx \le \lim_{n\to\infty}\int_0^1 \frac{B}{n}dx$$

Letting $n\to\infty$ then $\frac{1}{n}\to 0$

$$\int_0^1 0\,dx = 0$$

3. Find an example of a sequence of continuous functions on (0,1) that converges pointwise to a continuous function on (0.1), but the convergence is not uniform.

Choose $f_n(x) = \dfrac{1}{nx}$. We see that $f_n \xrightarrow{n\to\infty} 0$ for any individual x (thus pointwise). Testing for uniform convergence let $\varepsilon > 1$ and choose $x_n = \dfrac{1}{2n}$. Thus we get that

$$f_n(x_n) = \frac{1}{n\frac{1}{2n}} = 2 > 1 = \varepsilon$$

Thus for $\varepsilon = 1$ we get that there is no n such that $|f - f_n| < \varepsilon$ for all $x \in (0,1)$

4. True/False; prove or find a counterexample to the following statement: If $\{f_n\}$ is a sequence of everywhere discontinuous functions on $[0,1]$ that converge uniformly to a function f, then f is everywhere discontinuous.

False! Define a discontinuous everywhere function f_n

$$f_n(x) = \begin{cases} \frac{1}{n} & x \in \mathbb{Q} \cap [0,1] \\ 0 & x \notin \mathbb{Q} \cap [0,1] \end{cases}$$

Which as $n \to \infty$ we get $f_n \to f \equiv 0$. To test uniform convergence we get

$$\sup_{x \in [0,1]} [f_n(x) - f(x)] = \sup_{x \in [0,1]} \left[\frac{1}{n} - 0 \right] = \frac{1}{n} \xrightarrow{n \to \infty} 0$$

Thus we found a function that is discontinuous everywhere (because $\mathbb{Q}$ is dense in $\mathbb{R}$) and it converges uniformly.

5. Suppose $f : [0,1] \to \mathbb{R}$ is Riemann integrable. For the following two exercises define the number

$$||f||_{L^1} := \int_0^1 |f(x)|\, dx$$

Suppose that $\{f_n\}$ is a sequence of Riemann integrable functions on $[0,1]$ that converges uniformly to 0. Show that

$$\lim_{n \to \infty} ||f_n||_{L^1} = 0$$

Since $\{f_n\}$ converge uniformly on $[0,1]$. Thus for all $\varepsilon > 0$ there exists N such that for all $n \geq N$ then

$$|f_n| < \varepsilon$$

Meaning for sufficiently large n we have

$$\int_0^1 |f_n(x)|\, dx < \int_0^1 \varepsilon\, dx = \varepsilon$$

Thus letting $\varepsilon \to 0$ we get that

$$\lim_{n \to \infty} ||f_n||_{L^1} = 0$$

6. Find a sequence $\{f_n\}$ of Riemann integrable functions on [0,1] converging pointwise to 0, but

$$\lim_{n \to \infty} ||f_n||_{L^1} \textbf{ does not exist (is } \infty\textbf{)}$$

Define a function f_n

$$f_n(x) = \begin{cases} n^2 & 0 \leq x < \dfrac{1}{n} \\ 0 & \dfrac{1}{n} \leq x \leq 1 \end{cases}$$

We get that for all $n > \dfrac{1}{x}$ then $x_n = 0$. Evaluating the limit

$$\begin{aligned}
\lim_{n\to\infty} ||f_n||_{L^1} &= \lim_{n\to\infty} \int_0^1 f_n(x)\,dx \\
&= \lim_{n\to\infty} \int_0^{1/n} f_n(x)\,dx + \lim_{n\to\infty} \int_{1/n}^1 f_n(x)\,dx \\
&= \lim_{n\to\infty} \int_0^{1/n} n^2\,dx + \lim_{n\to\infty} \int_{1/n}^1 0\,dx \\
&= \lim_{n\to\infty} \int_0^{1/n} n^2\,dx + 0 \\
&= \lim_{n\to\infty} n^2 x\big|_{x=0}^{x=1/n} \\
&= \lim_{n\to\infty} n \\
&= \text{diverges to } \infty
\end{aligned}$$

7. We say that a sequence of functions $f_n : \mathbb{R} \to \mathbb{R}$ converges uniformly on compact subsets if for every $k \in \mathbb{N}$ the sequence f_n converges uniformly on $[-k, k]$. Prove that if $f_n : \mathbb{R} \to \mathbb{R}$ is a sequence of continuous functions converging uniformly on a compact subsets, then the limit is continuous.

Suppose $\{f_n\} \to f$. Then for all $\varepsilon > 0$ there exists some corresponding N_k such that for all $n \geq N_k$ we have

$$\sup_{x \in [-k,k]} |f_n(x) - f(x)| < \frac{\varepsilon}{3}$$

Since each f_n is continuous, then for all $\varepsilon > 0$ there exists $\delta_n > 0$ such that if $|\xi - x|$ then

$$|f_n(x) - f_n(\xi)| < \frac{\varepsilon}{3}$$

For all $x \in \mathbb{R}$

Fix $\varepsilon > 0$. Choose $\delta = \min\{\delta_n\}$. Fix arbitrary $\xi \in \mathbb{R}$ and we choose k such that $|\xi| \leq k$. Thus for any $\varepsilon > 0$ there exists a corresponding N_k such that for all $n \geq N_k$ AND if $|x - \xi| < \delta$ then we have

$$\begin{aligned} |f(\xi) - f(x)| &= |f(\xi) - f_n(\xi) + f_n(\xi) - f_n(x) + f_n(x) - f(x)| \\ &\leq |f(\xi) - f_n(\xi)| + |f_n(\xi) - f_n(x)| + |f_n(x) - f(x)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon \end{aligned}$$

Thus for all $\varepsilon > 0$ we have $\delta > 0$ such that if $|\xi - x| < \delta$ then

$$|f(\xi) - f(x)| < \varepsilon$$

For all $\xi \in \mathbb{R}$. Hence $f(x)$ is continuous.

8. Prove that if $f_n : \mathbb{R} \to \mathbb{R}$ is a sequence of functions Riemann integrable on any closed and bounded interval $[a, b]$, and converging uniformly on compact subset to an $f : \mathbb{R} \to \mathbb{R}$, then for any interval $[a, b]$ we have f is Riemann integrable on $[a, b]$, and

$$\int_a^b f(x)\,dx = \lim_{n \to \infty} \int_a^b f_n(x)\,dx$$

Since f_n converges uniformly on compact subsets, then we choose $k = \max\{|a|, |b|\}$. Hence we get that $[a, b] \subset [-k, k]$. Thus by problem 7, f is continuous and thus, Riemann integrable. Next, fix $\varepsilon > 0$. Since f_n converges uniformly to f we get that there exists N such that all $n \geq N$ we have

$$\sup_{x \in [a,b]} |f_n(x) - f(x)| < \frac{\varepsilon}{b - a}$$

Evaluating the following limit of the integral

$$\lim_{n\to\infty}\int_a^b |f_n(x)-f(x)|\,dx < \int_a^b \frac{\varepsilon}{b-a}\,dx = (b-a)\frac{\varepsilon}{b-a} = \varepsilon$$

Thus, by linearity of the integral

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\int_a^b f_n(x)\,dx$$

9. Find a sequence of Lipschitz continuous functions on $[0,1]$ whose uniform limit is $\sqrt{x}$, which is a non-Lipschitz function.

Recall that if each f_n is Lipschitz continuous on the set $[0,1]$, then there exists a constant K_n such that for all $x, y \in [0,1]$

$$|f_n(x)-f_n(y)| \le K_n|x-y| \quad \text{or} \quad \frac{|f_n(x)-f_n(y)|}{|x-y|} \le K_n$$

Define f_n as the following

$$f_n(x) = \sqrt{x+\frac{1}{n}} \quad \text{for} \quad K_n = \frac{\sqrt{n}}{2}$$

We can easily see that f_n converges pointwise to $\sqrt{x}$ because as $n\to\infty$ then $f_n(x)\to\sqrt{x}$. To show that f_n is Lipschitz, fix $x, y \in [0,1]$, thus we get that

$$\begin{aligned}
\frac{|f_n(x)-f_n(y)|}{|x-y|} &= \frac{\left|\sqrt{x+\frac{1}{n}}-\sqrt{y+\frac{1}{n}}\right|}{|x-y|} \\
&= \frac{\left|\sqrt{x+\frac{1}{n}}-\sqrt{y+\frac{1}{n}}\right|}{|x-y|}\cdot\frac{\left|\sqrt{x+\frac{1}{n}}+\sqrt{y+\frac{1}{n}}\right|}{\left|\sqrt{x+\frac{1}{n}}+\sqrt{y+\frac{1}{n}}\right|} \quad \text{(multiplying by the conjugate)} \\
&= \frac{\left|x+\frac{1}{n}-y-\frac{1}{n}\right|}{|x-y|\cdot\left|\sqrt{x+\frac{1}{n}}+\sqrt{y+\frac{1}{n}}\right|} \\
&= \frac{1}{\left|\sqrt{x+\frac{1}{n}}+\sqrt{y+\frac{1}{n}}\right|} \\
&\le \frac{1}{2\sqrt{\frac{1}{n}}} = \frac{\sqrt{n}}{2} = K_n
\end{aligned}$$

Thus every f_n is Lipschitz. To show uniform convergence, fix $\varepsilon > 0$ and choose $N > \frac{1}{2\varepsilon+\varepsilon^2}$. Thus for all $n > N$ we get

$$\begin{aligned}
\sup_{x\in[0,1]} \left\{ \sqrt{x+\frac{1}{n}} - \sqrt{x} \right\} &= \sqrt{1+\frac{1}{n}} - \sqrt{1} \qquad \text{(Since } \sqrt{x} \text{ is monotone)} \\
&< \sqrt{1+\frac{1}{\frac{1}{2\varepsilon+\varepsilon^2}}} - 1 \\
&= \sqrt{1+2\varepsilon+\varepsilon^2} - 1 \\
&= 1+\varepsilon-1 = \varepsilon
\end{aligned}$$

10 On the other hand, show that if $f_n : S \to \mathbb{R}$ are Lipschitz with a uniform constant K (Meaning all of them satisfy the definition with the same constant) and $\{f_n\}$ converges pointwise to $f : S \to \mathbb{R}$, then the limit f is a Lipschitz continuous function with Lipschitz constant K.

Since f_n is Lipschitz, we get that for any $x, y \in S$

$$\frac{|f_n(x) - f_n(y)|}{|x-y|} \leq K$$

Since f_n converges pointwise, it means for all $\varepsilon > 0$ there exists N' such that for all $n \geq N'$ we get

$$|f_n(x) - f(x)| < \varepsilon$$

for all $x \in S$
Fix $\varepsilon > 0$, Then choose $N = N'$ such that for all $n \geq N$, and for any $x, y \in S$

$$\begin{aligned}
\frac{|f(x)-f(y)|}{|x-y|} &= \frac{|f(x)-f_n(x)+f_n(x)-f_n(y)+f_n(y)-f(y)|}{|x-y|} \\
&\leq \frac{|f(x)-f_n(x)|+|f_n(x)-f_n(y)|+|f_n(y)-f(y)|}{|x-y|} \\
&< \frac{\varepsilon}{|x-y|} + K + \frac{\varepsilon}{|x-y|}
\end{aligned}$$

Letting $\varepsilon \to 0$ we get that

$$\frac{|f(x)-f(y)|}{|x-y|} \leq K$$

Homework 3

1. Find a closed form of the series

$$\sum_{k=2}^{\infty} kx^{k-2}$$

and the largest set on which the formula is valid

Re-indexing the sum by letting $k-2 \mapsto n$ or $k=2 \mapsto n=0$ we get

$$\sum_{n=0}^{\infty}(n+2)x^n$$

Splitting the sum

$$= \sum_{n=0}^{\infty} 2x^n + \sum_{n=0}^{\infty} nx^n$$

By geometric series we get that for $|x|<1$

$$= \frac{2}{1-x} + \frac{x}{(1-x)^2} = \frac{2-x}{(1-x)^2}$$

Thus we get he equality

$$\sum_{k=2}^{\infty} kx^{k-2} = \frac{2-x}{(1-x)^2}$$

When $|x|<1$ (The series diverges at $x=\pm 1$)

Let $f_n(x) := \frac{x}{1+(nx)^2}$. Notice that f_n are differentiable functions.

1. Show that $\{f_n\}$ converges uniformly to 0.

   Fix $\varepsilon > 0$, and choose $N > \frac{1}{2\varepsilon}$. Thus we for all $n \geq N$

   $$\begin{aligned}\sup_{x\in\mathbb{R}} \left| \frac{x}{1+(nx)^2} - 0 \right| &= \sup_{x\in\mathbb{R}} \frac{|x|}{1+(nx)^2} \\ &= \sup_{x\in\mathbb{R}_+} \frac{x}{1+(nx)^2}\end{aligned}$$

   By the graph of the function we see that $\frac{d}{dx} f_n(x) = 0 \iff x = \frac{1}{n}$. Thus we get that

   $$\begin{aligned}\sup_{x\in\mathbb{R}_+} \frac{x}{1+(nx)^2} &= \frac{\frac{1}{n}}{1+(n\frac{1}{n})^2} \\ &= \frac{1}{2n} \\ &< \frac{1}{2\frac{1}{2\varepsilon}} = \varepsilon\end{aligned}$$

Hence f_n converges uniformly to 0

2. Show that $|f_n'(x)| \leq 1$ for all x and all n

   Explicitly writing the derivative we get

$$|f_n'(x)| = \left| \frac{n^2x^2 - 1}{\left(n^2x^2 + 1\right)^2} \right|$$

   Let $a := nx$, it would suffice to show that $|a^2 - 1| \leq (a^2 + 1)^2$. If $a \geq 1$ then

$$\begin{aligned} a^2 - 1 &\leq (a^2 + 1)^2 \\ a^2 - 1 &\leq a^4 + 2a^2 + 1 \\ 0 &\leq a^4 + 3a^2 + 2 \end{aligned}$$

   If $0 \leq a \leq 1$ then

$$\begin{aligned} 1 - a^2 &\leq (a^2 + 1)^2 \\ 1 - a^2 &\leq a^4 + 2a^2 + 1 \\ 0 &\leq a^4 + 4a^2 \end{aligned}$$

   Thus we get that

$$|f_n'(x)| \leq 1$$

3. Show that $\{f_n'\}$ converges pointwise to a function discontinuous at the origin.

   For $x = 0$ we see that

$$f_n'(0) = -\frac{-1}{(1)^2} = 1$$

   Next fix a point $x \neq 0$ and claim that $\lim_{n\to\infty} f_n'(x) = 0$.Thus we get

$$\begin{aligned} \left| \frac{n^2x^2 - 1}{\left(n^2x^2 + 1\right)^2} - 0 \right| &< \frac{|n^2x^2 - 1|}{\left(n^2x^2\right)^2} && (1) \\ &\leq \frac{n^2x^2}{n^4x^4} + \frac{1}{n^4x^4} && (2) \\ &= \frac{1}{n^2x^2} + \frac{1}{n^4x^4} && (3) \\ & && (4) \end{aligned}$$

   Both $\frac{1}{n^2x^2}$ and $\frac{1}{n^4x^4}$ converge to 0 as $n \to \infty$ for $x \neq 0$. Thus the limit converges pointwise to

$$f(x) \begin{cases} 1, & x = 0 \\ 0, & x \neq 0 \end{cases}$$

3. Let $b > 0$,$b \neq 1$ be given

1. Show that for every $y > 0$ there exists a unique number x such that $y = b^x$. Define the logarithm base b, $\log_b : (0, \infty) \to \mathbb{R}$ by $\log_b(y) := x$

   Fix $b > 0$,$b \neq 0$. Recall that if $b > 1$ then b^x is strictly increasing and if $0 < b < 1$ then b^x is strictly decreasing. Thus in either case b^x is strictly monotone on $(0, \infty)$ and thus injective. Meaning if x existed, it would be unique. To show existence take $y > 0$. Since b^x is continuous (WLOG take $b > 1$) we get that

   $$\lim_{x \to -\infty} b^x = 0 < y < \infty = \lim_{x \to \infty} b^x$$

   By the intermediate value theorem, we get that for any y such that $0 < y < \infty$ there exists an x such that $-\infty < x < \infty$.

2. Show that $\log_b(x) = \frac{\ln x}{\ln b}$

   By part 1, there exists unique x such that for all $y > 0$ we have $y = b^x$ and thus $\log_b(y) = x$. Taking the natural log of both sides.

   $$y = b^x \iff \ln y = x \ln b$$

   Dividing both sides by $\ln b$ we get

   $$\frac{\ln y}{\ln b} = x$$

   Recall that $x = \log_b(y)$ and thus we get

   $$\frac{\ln y}{\ln b} = \log_b(y)$$

3. Prove that if $c > 0$, $c \neq 1$, then $\log_b(x) = \frac{\log_c(x)}{\log_c(b)}$

   Similar to the first problem we choose to take $\log_c$ of both sides.

4. Prove $\log_b(xy) = \log_b(x) + \log_b(y)$, and $\log_b(x^y) = y \log_b(x)$

   (I'm copy-pasting the solution from another homework which is in terms of a instead of b) Since we define $\log_a x$ as the inverse function of a^x we get that $\log_a(a^x) = x$ and $a^{\log_a x} = x$. Let $\log_a(x) + \log_a(y) = u$ for some real number u. Razing both sides over a and applying exponent rules we get.

   $$a^{\log_a(x) + \log_a(y)} = a^u$$
   $$a^{\log_a(x)} a^{\log_a(y)} = a^u$$
   $$xy = a^u$$

Taking the $\log_a$ of both sides

$$\log_a(xy) = \log_a a^u = u$$

Meaning

$$\log_a(x) + \log_a(y) = \log_a(xy)$$

Next, letting $y \log_a(x) = v$ for some real number v. Razing both sides over a and applying exponent rules we get.

$$a^{y \log_a(x)} = a^v$$

$$\left(a^{\log_a(x)}\right)^y = a^v$$

$$x^y = a^v$$

Taking the $\log_a$ of both sides

$$\log_a x^y = \log_a a^v = v$$

Meaning

$$y \log_a(x) = \log_a(x^y)$$

4. Using the logarithm find

$$\lim_{n\to\infty} n^{\frac{1}{n}}$$

Recall that for all $n > 0$ we see that $n^{\frac{1}{n}} > 0$. Thus we can take $e^{\ln}$ of the function and it won't change and thus we get.

$$\begin{aligned}
\lim_{n\to\infty} n^{\frac{1}{n}} &= \lim_{n\to\infty} e^{\ln n^{\frac{1}{n}}} \\
&= \lim_{n\to\infty} e^{\frac{\ln n}{n}} \\
&= e^0 \qquad \text{(linear growth is faster than logarithmic growth)} \\
&= 1
\end{aligned}$$

5. Use the geometric sum formula to show (for $t \neq -1$)

$$1 - t + t^2 - \cdots + (-1)^n t^n = \frac{1}{1+t} - \frac{(-1)^{n+1} t^{n+1}}{1+t}$$

Using this fact show

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

for all $x \in (-1, 1]$. Finally, find the limit of the alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

Recall the geometric series

$$\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r} \quad (r \neq 1)$$

We get that our sum for $r = -t$ or

$$\sum_{k=0}^{n} (-t)^k = \frac{1 - (-t)^{n+1}}{1 - (-t)} = \frac{1}{1+t} - \frac{(-1)^{n+1} t^{n+1}}{1+t} \quad (-t \neq 1 \text{ or } t \neq -1)$$

Next fix $|t| < 1$. Thus we get that

$$\lim_{n\to\infty} \sum_{k=0}^{n} (-t)^k = \frac{1}{1+t} - \lim_{n\to\infty} \frac{(-1)^{n+1} t^{n+1}}{1+t} = \frac{1}{1+t} - 0$$

Thus the infinite sum gets us

$$\sum_{k=0}^{\infty} (-t)^k = \frac{1}{1+t}$$

Integrating both sides with respect to t we get

$$\int_0^x \sum_{k=0}^{\infty} (-t)^k = \int_0^x \frac{1}{1+t}$$

$$\sum_{k=0}^{\infty} \frac{(-1)^k x^{k+1}}{x+1} - \sum_{k=0}^{\infty} \frac{(-1)^k 0^{k+1}}{0+1} = \ln(1+x) - \ln(1+0)$$

Simplifying and Re-indexing we get that for all $x \in (1, 1]$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

Letting $x = 1$ we get

$$\ln(2) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \cdots$$

6. Show that e^x is convex, in other words, show that if $a \leq x \leq b$, then

$$e^x \leq e^a \frac{b-x}{b-a} + e^b \frac{x-a}{b-a}$$

We will generalize for any function f that is twice differentiable on (a, b) and $f''(x) \geq 0$ for all $x \in (a, b)$. First fix $a < x < b$. Next define a function g such that

$$g(x) := f(x) - f(a)\frac{b-x}{b-a} - f(b)\frac{x-a}{b-a}$$

It would suffice to show that $g(x) \leq 0$ for all $x \in (a, b)$ because

$$g(x) \leq 0 \implies f(x) - f(a)\frac{b-x}{b-a} - f(b)\frac{x-a}{b-a} \leq 0$$

The endpoint values give us $g(a) = 0$ and $g(b) = 0$. Evaluating the 2nd derivative of $g(x)$ gives us

$$g''(x) = f''(x) \geq 0$$

Meaning that $g'(x)$ is monotone increasing. Next fix $x \in (a, b)$. Applying the mean value theorem to the interval $[a, x]$ we get that there exists $u \in [a, x]$ such that

$$g'(u) = \frac{g(x) - g(a)}{x - a} = \frac{g(x)}{x - a}$$

Similarly applying the mean value theorem to the interval $[x, b]$ we get that there exists $v \in [x, b]$ such that

$$g'(u) = \frac{g(b) - g(x)}{b - x} = \frac{-g(x)}{b - x}$$

Using the monotonicity of g' we see that $u \leq v \implies g'(u) \leq g'(v)$ Substituting the MVT formulas

$$\frac{g(x)}{x-a} \leq \frac{-g(x)}{b-x}$$

$$g(x)(b-x) \leq -g(x)(x-a)$$

$$g(x)(b-x+x-a) \leq 0$$

$$g(x)(b-a) \leq 0$$

$$g(x) \leq 0 \quad \text{(because } a < b\text{)}$$

Thus f is convex.

Homework 4

Let X be the set of continuous functions on $[0,1]$. Let $\varphi : [0,1] \to (0,\infty)$ be continuous. Define

$$d(f,g) := \int_0^1 |f(x) - g(x)|\, \varphi(x)\, dx$$

Show that (X,d) is a metric space.

We will prove (X,d) is a metric space axiomatically

1. **Identity** $d(f,f) = 0$
   Choose arbitrary $f \in X$ then $d(f,f)$ gives us

   $$\int_0^1 |f(x) - f(x)|\, \varphi(x)\, dx$$
   $$\int_0^1 |0|\, \varphi(x)\, dx$$
   $$\int_0^1 0\, dx = 0$$

   Thus the identity holds

2. **Positivity** $d(f,g) > 0 \iff f \neq g$
   Since $\varphi > 0$ then it would suffice to show that

   $$\int_0^1 |f(x) - g(x)|\, dx > 0$$

   because the product of two positive values is positive. Suppose $f \neq g$. Assume for contradiction that

   $$\int_0^1 |f(x) - g(x)|\, dx = 0$$

   However this means that $|f - g| = 0$ for all $x \in [0,1]$ (or f and g differ by finitely many values of x but then continuity of g or f fails). Thus $f = g$ a contradiction. Therefore

   $$\int_0^1 |f(x) - g(x)|\, dx > 0$$

   Thus positivity holds

3. **Symmetry** $d(f,g) = d(g,f)$

$$\begin{aligned} d(f,g) &= \int_0^1 |f(x) - g(x)|\, \varphi(x)\, dx \\ &= \int_0^1 |(-1)(g(x) - f(x))|\, \varphi(x)\, dx \\ &= \int_0^1 |(g(x) - f(x))|\, \varphi(x)\, dx = d(g,f) \end{aligned}$$

Thus symmetry holds

4. Triangle inequality $d(f,h) \leq d(f,g) + d(g,h)$ Let $f, g, h \in X$. Then we get

$$d(f,h) = \int_0^1 |f(x) - h(x)|\, \varphi(x)\, dx$$

Using triangle inequality of the absolute value we get

$$\int_0^1 |f(x) - h(x)|\, \varphi(x)\, dx \leq \int_0^1 (|f(x) - g(x)| + |g(x) - h(x)|)\, \varphi(x)\, dx$$

Distributing $\varphi(x)\, dx$ and using linearity of the integral we get

$$\int_0^1 |f(x) - g(x)|\, \varphi(x)\, dx + \int_0^1 |g(x) - h(x)|\, \varphi(x)\, dx = d(f,g) + d(g,h)$$

Thus the triangle inequality holds

2. Let $C^1([a,b],\mathbb{R})$ be the set of once continuously differentiable functions on $[a,b]$. Define

$$d(f,g) := ||f - g||_u + ||f' - g'||_u$$

where $||\cdot||_u$ is the uniform norm. Prove that d is a metric.

Recall the definition of the uniform norm that for some function $h : [a,b] \to \mathbb{R}$, then

$$||h(x)||_u := \sup_{x \in [a,b]} |h(x)|$$

We will prove d is a metric on $C^1([a,b],\mathbb{R})$ space axiomatically.

1. **Identity** $d(f,f) = 0$
For any function f we see, that

$$\sup_{x \in [a,b]} |f(x) - f(x)| = \sup 0 = 0 \quad \text{and} \quad \sup_{x \in [a,b]} |f'(x) - f'(x)| = \sup 0 = 0$$

2. **Positivity** $d(f,g) > 0 \iff f \neq g$
   Suppose $f \neq g$. Assume for contradiction that

$$\sup_{x\in[a,b]} |f(x) - g(x)| = 0 \quad \text{and} \quad \sup_{x\in[a,b]} |f'(x) - g'(x)| = 0$$

   However this means that $|f - g| = 0$ for all $x \in [0,1]$. Thus $f = g$ a contradiction.

3. **Symmetry** $d(f,g) = d(g,f)$
   Using symmetry of the absolute value we get that

$$\sup_{x\in[a,b]} |f(x) - g(x)| = \sup_{x\in[a,b]} |g(x) - f(x)|$$

$$\sup_{x\in[a,b]} |f'(x) - g'(x)| = \sup_{x\in[a,b]} |g'(x) - f'(x)|$$

4. Triangle inequality $d(f,h) \leq d(f,g) + d(g,h)$
   Using triangle inequality of the absolute value

$$|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)|$$

   Taking the supremum over $x \in [a,b]$ we get

$$\sup_{x\in[a,b]} |f(x) - h(x)| \leq \sup_{x\in[a,b]} |f(x) - g(x)| + \sup_{x\in[a,b]} |g(x) - h(x)|$$

   Similarly starting with $|f' - h'|$ we get

$$\sup_{x\in[a,b]} |f'(x) - h'(x)| \leq \sup_{x\in[a,b]} |f'(x) - g'(x)| + \sup_{x\in[a,b]} |g('x) - h'(x)|$$

3. Let X be a set and d_1 and d_2 be two metrics on X. Suppose that there exists an $\alpha > 0$ and $\beta > 0$ such that

$$\alpha d_1(x,y) \leq d_2(x,y) \leq \beta d_1(x,y) \quad \textbf{for all } x, y \in X$$

Show that U is open in (X, d_1) if and only if U is open in (X, d_2). That is the topologies of (X, d_1) and (X, d_2) are the same.

Suppose that U is open in (X, d_1). By the the definition of an open set. For all $y \in X$. there exists $\varepsilon > 0$ such that

$$B(y,\varepsilon) = \left\{ x : d_1(x,y) < \frac{\varepsilon}{\beta} \right\} \subseteq U$$

Recall that $d_2(x,y) \leq \beta d_1(x,y)$. Thus we get

$$d_2(x,y)\frac{1}{\beta} < \frac{\varepsilon}{\beta}$$

Thus for the same y there exists $\varepsilon > 0$ such that

$$B(y, \varepsilon) = \{x : d_2(x, y) < \varepsilon\} \subseteq U$$

And thus U is open in (X, d_2). To prove the converse suppose that U is open in (X, d_2). By the the definition of an open set. For all $y \in X$. there exists $\varepsilon > 0$ such that

$$B(y, \varepsilon) = \{x : d_2(x, y) < \alpha\varepsilon\} \subseteq U$$

Recall that $\alpha d_1(x, y) \leq d_2(x, y) < \alpha\varepsilon$. Thus we get

$$d_1(x, y)\frac{1}{\beta} < \frac{\varepsilon}{\beta}$$

Thus for the same y there exists $\varepsilon > 0$ such that

$$B(y, \varepsilon) = \{x : d_1(x, y) < \varepsilon\} \subseteq U$$

And thus U is open in (X, d_2).

4. Let (X, d) be a metric space.

(a) For any $x \in X$ and $\delta > 0$ show $\overline{B(x, \delta)} \subset C(x, \delta)$

(b) Is it always true that $\overline{B(x, \delta)} = C(x, \delta)$? Prove or find a counter example.

Recall the definitions

$$B(x, \delta) = \{y \in X : d(x, y) < \delta\} \quad C(x, \delta) = \{y \in X : d(x, y) \leq \delta\}$$

Where $\overline{B(x, \delta)}$ is the closure of $B(x, \delta)$

(a) Let $y \in \overline{B(x, \delta)}$ be arbitrary. Assume for contradiction that $d(x, y) > \delta$. Define

$$\varepsilon := \frac{d(x, y) - \delta}{2} > 0$$

Take $z \in B(y, \varepsilon)$. By the inverse triangle inequality we get

$$d(x, z) \geq d(x, y) + d(y, z) > d(x, y) - \varepsilon$$

$$= d(x, y) - \frac{d(x, y) - \delta}{2} = \frac{d(x, y) + \delta}{2} > \frac{2\delta}{2} = \delta$$

Thus we see that $d(x, z) > \delta$ meaning $z \notin B(x, \delta)$. Hence

$$B(y, \varepsilon) \cap B(x, \delta) = \varnothing$$

Contradicting that $y \in \overline{B(x, \delta)}$. Hence $d(x, y) \leq \delta$ and thus $\overline{B(x, \delta)} \subset C(x, \delta)$.

(b) Define $d(x, y)$ such that

$$d(x,y) = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases}$$

Fix $x \in X$ and $\delta = 1$. Thus the open ball is

$$B(x, 1) = \{y \in X : d(x, y) < 1\} = \{x\}$$

Since the ball is a single ton, the open set is also a closed set and thus it's own closure. However, computing the closed ball

$$C(x, \delta) = \{y \in X : d(x, y) \leq 1\} = X$$

Thus the closed ball and the closure are not equal.

5. For every $x \in \mathbb{R}^n$ and every $\delta > 0$ define a rectangle

$$R(x, \delta) := (x_1 - \delta, x_1 + \delta) \times (x_2 - \delta, x_2 + \delta) \times \cdots \times (x_n - \delta, x_n + \delta)$$

Show that these sets generate the same open sets as the balls in a standard metric. that is, show that a set $U \subset \mathbb{R}^n$ is open in the sense of the standard metric if and only if for every point $x \in U$ there exists $\delta > 0$ such that $R(x, \delta) \subset U$

Suppose that U is open in $\mathbb{R}^n$. Then for every $x \in U$ there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset U$. Next for any given rectangle $R(x, \delta)$ the diagonal of the set must be

$$\sqrt{[(x_1 - \delta) - (x_1 + \delta)]^2 + [(x_2 - \delta) - (x_2 + \delta)]^2 + \cdots + [(x_3 - \delta) - (x_3 + \delta)]^2}$$

$$= \sqrt{4n\delta^2} = 2\delta\sqrt{n}$$

If we choose $\delta = \dfrac{\varepsilon}{2\sqrt{n}}$, then for every point y in $R(x, \delta)$ we get

$$d(x, y) < 2\delta\sqrt{n} = \frac{\varepsilon}{2\sqrt{n}} 2\sqrt{n} = \varepsilon$$

Thus the rectangle is a subset of the open ball around x. Hence for all $x \in U$ we choose $\delta > 0$ such that $R(x, \delta) \subset U$. For the converse suppose that for all $x \in U$, then there exists $\delta > 0$ such that $R(x, \delta) \subset U$. If we choose $\varepsilon = \delta$ then the ε-ball is inscribed in the δ-rectangle and thus

$$B(x, \varepsilon) \subset R(x, \delta) \subset U$$

Hence for all $x \in U$ we choose $\varepsilon > 0$ such that $B(x, \varepsilon) \subset U$.

6. Suppose $\{x_n\}_{n=1}^{\infty}$ converges to x. Suppose $f : \mathbb{N} \to \mathbb{N}$ is a one-to-one function. Show that $\{x_{f(n)}\}_{n=1}^{\infty}$ converges to x.

Fix $\varepsilon > 0$. Since $\{x_n\}_{n=1}^{\infty}$ converges, there exists N such that for all $n > N$, then $d(x_n, x) < \varepsilon$. Define the set A as

$$A := \{n \in \mathbb{N} : f(n) < N\}$$

Since f is injective, then there are $N-1$ distinct numbers such that $f(n) \in A$, so $|A| \leq N-1$ and thus is finite. Therefore if we choose $N_0 = 1 + \max\{A, 1\}$ then for all $n > N_0$ we have $d(x_{f(n)}, x) < \varepsilon$. Thus $\{x_{f(n)}\}_{n=1}^{\infty}$ converges to x

7. A set $S \subset X$ is said to be *dense* in X if $X \subset \overline{S}$. Or in other words exists a sequence $\{x_n\} \in S$ that converges to x. Prove that $\mathbb{R}^n$ contains a countable dense subset.

First we will prove that $\mathbb{Q}^n$ is countable for all finite n. Then we will prove that $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$. Since $\mathbb{Q}$ is countable, there exists a surjective function $f : \mathbb{N} \to \mathbb{Q}$. Next we will define a function

$$F : \mathbb{N} \times \mathbb{N} \to \mathbb{Q} \to \mathbb{Q} \quad F(a, b) = (f(a), f(b))$$

Thus we get F is surjective as well. Next, the inverse of the Cantor Pairing function

$$\pi^{-1} : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$$

is at least surjective (bijective) and composition of surjective functions is surjective. Composing $F \circ \pi^{-1}$ the functions we get a surjective function and thus $\mathbb{Q}^2$ is countable. Next suppose that $\mathbb{Q}^k$ is countable. Then using the same method as before we get that $\mathbb{Q}^k \times \mathbb{Q}$ is also countable. Thus by induction for all n, $\mathbb{Q}^n$ is countable.

Next let's prove that $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$. Recall that $\mathbb{Q}$ is dense in $\mathbb{R}$ (see proof in another homework). Thus it would suffice to show that for any $x \in \mathbb{R}^n$, there exists $q \in \mathbb{Q}^n$ such that

$$||x - q|| < \varepsilon$$

. Using coordinate-wise approximation we can choose that for any coordinate of $x = (x_1, x_2, \cdots, x_n)$ then we choose q_i such that

$$|x_i - q_i| < \frac{\varepsilon}{\sqrt{n}}$$

Therefore we get a point $q = (q_1, q_2, \cdots, q_n)$ such that

$$||x - q|| = \sqrt{|x_1 - q_1|^2 + |x_2 - q_2|^2 + \cdots + |x_n - q_n|^2}$$

$$< \sqrt{\left(\frac{\varepsilon}{\sqrt{n}}\right)^2 + \left(\frac{\varepsilon}{\sqrt{n}}\right)^2 + \cdots + \left(\frac{\varepsilon}{\sqrt{n}}\right)^2} = \sqrt{n\frac{\varepsilon^2}{n}} = \varepsilon$$

Hence $||x - q|| < \varepsilon$ and thus $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$

Homework 5

1. Prove the general Bolzano-Weierstrass theorem: Any bounded sequence $\{x_k\}$ in $\mathbb{R}^n$ has a convergent subsequence.

Since the sequence $\{x_k\} \subset \mathbb{R}^n$, There is $M < \infty$ such that $||x_k|| < M$ for all k. Define the cube Q_0

$$Q_0 := [-M, M]^n = \prod_{i=i}^{n} [-M, M]$$

Next we will conduct a recursive split of the cube. For any closed rectangle

$$Q_k = \prod_{i=i}^{n} [(a_k)_i, (b_k)_i]$$

Let the midpoint of each coordinate be

$$(m_k)_i = \frac{(b_k)_i - (a_k)_i}{2}$$

This splits each coordinate interval $[(a_k)_i, (b_k)_i]$ into two sub intervals $[(a_k)_i, (m_k)_i] + [(m_k)_i, (b_k)_i]$, hence 2^n sub rectangles total. By the Pigeonhole principle, since Q_k contains infinitely many points of the sequence $\{x_n\}$, and 2^n sub-regions. At least one of the regions must have infinitely many points. We will define this region as Q_{k+1}. We get the nested chain

$$Q_0 \supset Q_1 \supset Q_2 \supset$$

With each Q_k containing infinitely many points. Each coordinate interval length halves each step.

$$\left| \frac{(b_k)_i - (a_k)_i}{2} \right| = |(b_{k+1})_i - (a_{k+1})_i|$$

If we fix $\varepsilon > 0$ and choose $K = \log_2(M \frac{1}{\varepsilon})$ then for all $k > K$ we get that

$$\begin{aligned} |(b_K)_i - (a_K)_i| &= \frac{M}{2^K} \\ &< \frac{M}{2^{\log_2(M\frac{1}{\varepsilon})}} = \frac{M}{\frac{M}{\varepsilon}} D = \varepsilon \end{aligned}$$

$$|(b_k)_i - (a_k)_i| = \frac{M}{2^k} < \varepsilon$$

This each axis cube goes to zero. Because each Q_k contains infinitely many terms we can choose strictly increasing indices x_m. Choose m_1 such that $x_{m_1} \in Q_1$. Having chosen m_k, choose m_{k+1} such that $x_{m_{k+1}} \in Q_{k+1}$. Thus we have a subsequence x_m. To show that it converges to a point $x := (x_1, x_2, \cdots x_n)$, recall that each coordinate interval of Q_k goes to zero and each index of x lies in the intersection of all the closed cubes. By the squeeze theorem, we get that each index of x_k converges to each index of x. Thus the bounded infinite sequence in $\mathbb{R}^n$ has a convergent subsequence.

2. Prove the following version of the intermediate value theorem. Let (X, d) be a connected metric space and $f : X \to \mathbb{R}$ a continuous function. Suppose that there exists $x_0, x_1 \in X$ and $y \in \mathbb{R}$ such that $f(x_0) < y < f(x_1)$. Then prove that there exists a $z \in X$ such that $f(z) = y$

Suppose that $f(X) \subset \mathbb{R}$ was not connected. Then there would exist two non-empty, open sets $U, V \subset \mathbb{R}$ such that

$$f(X) = U \cup V \qquad U \cap V = \varnothing$$

Since f is continuous, the preimage $f^{-1}(U)$ and $f^{-1}(V)$ are both open and thus also disjoint. Contradicting connectedness of X. Therefore $f(X)$ is connected. Similarly, we can prove that any connected subset of $\mathbb{R}$ is an interval. Let A be a connected subset of $\mathbb{R}$ and $a, b \in A$ where $a < b$. Fix an arbitrary t such that $a < t < b$. Assume for contradiction that $t \notin A$. Then we can define two nonempty sets U, V

$$U := (-\infty, t) \cap A \qquad V := (t, \infty) \cap A$$

However $U \cap V = \varnothing$, by definition contradiction connectedness of A. Therefore any connected subset of $\mathbb{R}$ must contain every number between any two elements, an interval. Since we're, given $f(x_0) < y < f(x_1)$, and $f(X)$ is a connected interval, then y is in the image of $f(X)$. Thus, there exists $z \in X$ such that $f(z) = y$.

3. Take the metric space of continuous functions $C([0, 1], \mathbb{R})$. Let $k : [0, 1] \times [0, 1] \to \mathbb{R}$ be a continuous function. Given $f \in C([0, 1], \mathbb{R})$ define

$$\varphi_f(x) := \int_0^1 k(x, y) f(y)\, dy$$

(a) Show that $T(f) := \varphi_f$ defines a function $T : C([0, 1], \mathbb{R}) \to C([0, 1], \mathbb{R})$

(b) Show that T is continuous

(a) Let $f \in C([0, 1], \mathbb{R})$ be arbitrary. Since k is continuous on $[0, 1]^2$ and f is continuous on $[0, 1]$ then the product is also continuous on $[0, 1]^2$, a compact set. Thus $g(x, y) := k(x, y)f(y)$ is uniformly continuous. Thus for all $\varepsilon > 0$ there exists $\delta > 0$ such that for all (x, y) and $(x', y') \in [0, 1]^2$

$$\sqrt{(x - x')^2 + (y - y')^2} < \delta \implies |g(x, y) - g(x', y')| < \varepsilon$$

Fix $\varepsilon > 0$ and take δ from uniform continuity. We get that for some fixed y, then

$$\sqrt{(x - x')^2 + (y - y)^2} = \sqrt{(x - x')^2} < \delta \implies |g(x, y) - g(x', y)| < \varepsilon$$

Thus $g(x,y)$ is continuous over $x \in [0,1]$. Since integration preserves continuity, we can also say that $\varphi_f(x)$ is continuous over all $x \in [0,1]$ or

$$T : C([0,1],\mathbb{R}) \to C([0,1],\mathbb{R})$$

(b) Define a constant M such that

$$M := \sup_{x,y\in[0,1]} k(x,y)$$

Suppose there was a sequence of functions $\{f_n\} \subset C([0,1],\mathbb{R})$ that converged uniformly to f. To show $T(f)$ is continuous we need to show that $T(f_n) \to T(f)$. Since $\{f_n\}$ is uniformly convergent, then for all $\varepsilon > 0$ there exists N such that for all $n > N$, then $||f_n(x) - f(x)|| < \dfrac{\varepsilon}{M}$. Recall that $T(f_n)$ is defined as

$$T(f_n(x)) = \int_0^1 k(x,y) f_n(y)\, dy$$

Subtract $T(f_n)$ from $T(f)$ and combine the integrals

$$|T(f_n(x)) - T(f(x))| = \left| \int_0^1 k(x,y)\left[f_n(y) - f(y)\right]\, dy \right|$$

By the integral triangle inequality

$$\begin{aligned} |T(f_n(x)) - T(f(x))| &\le \int_0^1 |k(x,y)\left[f_n(y) - f(y)\right]\, dy| \\ &= \int_0^1 |k(x,y)|\,|f_n(y) - f(y)|\, dy \end{aligned}$$

Use N from uniform convergence such that $|f_n(y) - f(y)| < \dfrac{\varepsilon}{M}$ and substitute $k(x,y)$ with M

$$\begin{aligned} &< \int_0^1 M \frac{\varepsilon}{M}\, dy \\ &= \varepsilon \end{aligned}$$

Hence the sequence $T(f_n)$ approaches $T(f)$ and thus T is continuous

4. Suppose $\gamma : (-1,1) \to \mathbb{R}^n$ and $\alpha \to \mathbb{R}^n$ be two differentiable curves such that $\gamma(0) = \alpha(0)$ and $\gamma'(0) = \alpha'(0)$. Suppose $F : \mathbb{R}^n \to \mathbb{R}$ is a differentiable function. Show that

$$\left.\frac{d}{dt}\right|_{t=0} F(\gamma(t)) = \left.\frac{d}{dt}\right|_{t=0} F(\alpha(t))$$

Since F is differentiable define it's derivative to be f. Where $\frac{d}{dx}F(x) = f(x)$. Then evaluating the derivative of $F(\gamma(t))$, using the chain rule we get

$$\frac{d}{dt}F(\gamma(t)) = F'(\gamma(t))\gamma'(t)$$

Evaluating at $t = 0$ we get

$$F'(\gamma(0))\gamma'(0)$$

Substituting $\gamma(0) = \alpha(0)$ and $\gamma'(0) = \alpha'(0)$

$$F'(\alpha(0))\alpha'(0)$$

Next evaluate the derivative of $F(\alpha(t))$, using the chain rule again and evaluating at $t = 0$

$$\frac{d}{dt}F(\alpha(t)) = F'(\alpha(t))\alpha'(t)$$

$$F'(\alpha(0))\alpha'(0)$$

Thus all we need to show that

$$\frac{d}{dt}F(\gamma(t)) = F'(\gamma(t))\gamma'(t)$$

Would be to prove the chain rule. (Which I did in another homework assignment)

5. Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ is differentiable and suppose that whenever $x^2 + y^2 = 1$, then $f(x, y) = 0$. Prove that there exists at least one point (x_0, y_0) such that $\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$

Fix some arbitrary $y \in (-1, 1)$. Thus $|y| < 1$ and therefore there exists exactly two real solutions x_1, x_2 that satisfy $x^2 = 1 - y^2$ or when $f(x, y) = 0$ Thus, by the mean value theorem there exists at least one point $c_1 \in [x_1, x_2]$ such that

$$\frac{\partial f}{\partial x}(c_1, y) = 0$$

For any arbitrary $y \in (0, 1)$ and there must be a continuous path from $(0, -1)$ to $(0, 1)$ of points that satisfy. Similarly, if we fix $|x| < 1$ there exists there exists exactly two real solutions y_1, y_2 that satisfy $y^2 = 1 - x^2$ Thus, by the mean value theorem there exists at least one point $c_2 \in [y_1, y_2]$ such that

$$\frac{\partial f}{\partial y}(x, c_2) = 0$$

For any arbitrary $x \in (0, 1)$ and there must be a continuous path from $(-1, 0)$ to $(1, 0)$ of points that satisfy. Take the intersection of these paths and you have a point (x_0, y_0) where

$$\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$$

Homework 6

1. Define $f : \mathbb{R}^2 \to \mathbb{R}$ as

$$f(x,y) = \begin{cases} (x^2+y^2)\sin\left((x^2+y^2)^{-1}\right) & \textbf{if } (x,y) \neq (0,0) \\ 0 & \textbf{else} \end{cases}$$

Show that f is differentiable at the origin but that it's not continuously differentiable.

Recall that if $f : \mathbb{R}^2 \to \mathbb{R}$ is differentiable at a, then there exists a linear map, $A \in (\mathbb{R}^2, \mathbb{R})$ such that

$$\lim_{h\to 0} \frac{||f(h+a) - f(a) - Ah||}{||h||} = 0$$

For $h = (x,y)$, write $r = ||h|| = \sqrt{x^2+y^2}$. For $h \neq (0,0)$

$$f(x,y) = (x^2+y^2)\sin\left(\frac{1}{x^2+y^2}\right) = r^2\sin\left(\frac{1}{r^2}\right)$$

Choose $A(h) \equiv 0$. Thus our limit becomes

$$\lim_{h\to 0} \frac{||f(h+a) - f(a) - Ah||}{||h||} = \lim_{r\to 0} \frac{|f(x,y)|}{r}$$

Using $|\sin| \leq 1$

$$|f(x,y)| = \left|r^2\sin\left(\frac{1}{r}\right)\right| \leq r^2$$

Thus

$$\lim_{r\to 0} \frac{|f(x,y)|}{r} \leq \lim_{r\to 0} \frac{r^2}{r} = \lim_{r\to 0} r = 0$$

Since the limit exists, then f is differentiable at the origin. To show that f is not C^1 at $(0,0)$, compute the partial derivative for $(x,y) \neq (0,0)$. Let $r = x^2+y^2$. Then $f = r\sin(1/r)$. Using the chain rule

$$\frac{\partial f}{\partial x} = \frac{\partial r}{\partial x}\sin\left(\frac{1}{r}\right) + r\cos\left(\frac{1}{r}\right)\frac{\partial}{\partial x}\frac{1}{r}$$

Where $\frac{\partial r}{\partial x} = 2x$ and $\frac{\partial}{\partial x}\frac{1}{r} = -\frac{1}{r^2}2x$. So it evaluates to

$$\frac{\partial f}{\partial x} = 2x\sin\left(\frac{1}{r}\right) - \cos\left(\frac{1}{r}\right)\frac{2x}{r}$$

Looking along the x-axis: $y = 0$, $r = x^2$ for $x \neq 0$ we get

$$\frac{\partial}{\partial x} f(x,0) = 2x\sin\left(\frac{1}{x^2}\right) - \cos\left(\frac{1}{x^2}\right)\frac{2}{x}$$

If f were continuously differentiable at $(x, y) = (0, 0)$, then $f_x(x, y) \to f(0, 0) = 0$ as $(x, y) \to 0$. Choose the sequence

$$x_n = \frac{1}{\sqrt{2\pi n}} \to 0$$

Thus $\sin\left(\frac{1}{x_n^2}\right) = 0$ and $\cos\left(\frac{1}{x_n^2}\right) = 1$ Thus as $x_n \to 0$

$$f(x_n, 0) = 2x_n \cdot 0 - 1 \cdot \sqrt{2\pi n} \to -\infty$$

Thus f_x does not tend to 0 as $x \to 0$ and thus f_x is discontinuous at $(0, 0)$

2. Let $B(0, 1) \subset \mathbb{R}^2$ be the unit ball, that is, the set given by $x^2 + y^2 < 1$. Suppose $f : B(0, 1) \to \mathbb{R}$ is a differentiable function such that $|f(0, 0)| \leq 1$, and $\left|\frac{\partial f}{\partial x}\right| \leq 1$ and $\left|\frac{\partial f}{\partial y}\right| \leq 1$ for all points in $B(0, 1)$

1. Find an $M \in \mathbb{R}$ such that $||f'(x, y)|| \leq M$ for all $(x, y) \in B(0, 1)$
2. Find $B \in \mathbb{R}$ such that $|f(x, y)| \leq$ for all $(x, y) \in B(0, 1)$

1. For any $(x_0, y_0) \in B(0, 1)$

   $$||f'(x_0, y_0)|| = |f_x(x, y)x_0 + f_y(x, , y)_0| \leq \sqrt{f_x(x, y)^2 + f_y(x, y)^2}\sqrt{x_0^2 + y_0^2}$$

   Since $\left|\frac{\partial f}{\partial x}\right| \leq 1$ and $\left|\frac{\partial f}{\partial y}\right| \leq 1$, we get that

   $$\sqrt{f_x(x, y)^2 + f_y(x, y)^2} \leq \sqrt{1 + 1} = \sqrt{2}$$

   Thus we can take $M = \sqrt{2}$

2. Fix (x, y) and consider the line segment from $(0, 0)$ to (x, y)

   $$\gamma(t) = (tx, ty), \quad t \in [0, 1], \quad g(t) := f(\gamma(t)) = f(tx, ty)$$

   Thus g is differentiable, using the chain rule

   $$g'(t) = f'(tx, ty)(x, y)$$

   Hence

   $$|g'(t)| = ||f'(tx, ty)(x, y)|| \leq M|(x, y)| \leq \sqrt{2}\sqrt{x^2 + y^2}$$

   Integrating from 0 to 1

   $$|f(x, y) - f(0, 0)| = g(1) - g(0)| \leq \int_0^1 |g'(t)|\, dt \leq \int_0^1 \sqrt{2}\sqrt{x^2 + y^2}\, dt = \sqrt{2}\sqrt{x^2 + y^2}$$

   Thus

   $$|f(x, y)| \leq |f(0, 0)| + \sqrt{2}\sqrt{x^2 + y^2}$$

   By definition of the unit disc, $\sqrt{x^2 + y^2} < 1$

   $$|f(0, 0)| + \sqrt{2}\sqrt{x^2 + y^2} \leq 1 + \sqrt{2}$$

   So we take $B = 1 + \sqrt{2}$

3. Suppose $f : \mathbb{R}^n \to \mathbb{R}$ and $h : \mathbb{R}^n \to \mathbb{R}$ are two differentiable function such that $f'(x) = h'(x)$ for all $x \in \mathbb{R}^n$. Prove that if $f(0) = h(0)$ then $f(x) = h(x)$ for all $x \in \mathbb{R}^n$

Define a function

$$g(x) := f(x) - h(x)$$

The derivative is given by

$$g'(x) = f'(x) - h'(x) = 0$$

So $g'(x)$ is the zero map from $\mathbb{R}^n \to \mathbb{R}$ Choose some arbitrary $x \in \mathbb{R}^n$ and consider the function $\gamma : [0, 1] \to \mathbb{R}$

$$\gamma(t) := g(tx)$$

By the chain rule

$$\gamma'(t) = g'(tx)[x]$$

But $g(tx) = 0$ for all t. Hence $\gamma'(t) = 0$ and thus is constant on $[0, 1]$. Thus

$$g(x) = \gamma(1) = \gamma(0) = g(0) = 0$$

Since x was arbitrary, $g(x) = 0$ for all $x \in \mathbb{R}^n$ and thus

$$f(x) = h(x)$$

4. Define $f : \mathbb{R}^2 \to \mathbb{R}^2 \backslash \{(0, 0)\}$ by $f(x, y) := (e^x \cos(y), e^x \sin(y))$.

1. Show that f is onto
2. Show that f' is invertible at all points
3. Show that f is not one-to-one, in fact, for every $(a, b) \in \mathbb{R}^2 \backslash \{(0, 0)\}$, there exists infinitely many different points $(x, y) \in \mathbb{R}^2$ such that $f(x, y) = (a, y)$

Consider the modulus of the function,

$$|f(x, y)| = \sqrt{e^{2x} \cos^2(y) + e^{2x} \sin^2(y)} = \sqrt{e^{2x}(\cos^2(y) + \sin^2(y))} = e^x$$

Which is entirely dependent on x. The angle of a point $f(x, y)$ that is made with the real x-axis is given as the following

$$\arctan\left(\frac{e^x \sin(y)}{e^x \cos(y)}\right) = \arctan\left(\tan(y)\right) = y$$

Which is entirely dependent on $y + 2\pi n$ for $n \in \mathbb{Z}$ (since $\cos(y + 2\pi n) = \cos y$ and $\sin(y + 2\pi n) = \sin y$)

1. To show that f is onto, let $(a,b) \in \mathbb{R}^2 \backslash \{(0,0)\}$ be arbitrary. The modulus of (a,b) is
$$|a,b| = \sqrt{a^2+b^2}$$
Since $(a,b) \neq (0,0)$ this forces $|(a,b)| > 0$ which shares a co-domain with e^x. and the angle is $\arctan\left(\frac{b}{a}\right)$. Thus we choose x such that
$$x = \log\left(\sqrt{a^2+b^2}\right)$$
and choose y such that
$$y = \arctan\left(\frac{b}{a}\right)$$
Thus we chose x, y such that $f(x,y) = (a,b)$. Thus f is onto

2. Compute the Jacobian matrix $Df(x,y)$
$$f'(x,y) = \begin{bmatrix} \frac{\partial}{\partial x}e^x \cos y & \frac{\partial}{\partial y}e^x \cos y \\ \frac{\partial}{\partial x}e^x \sin y & \frac{\partial}{\partial y}e^x \sin y \end{bmatrix} = \begin{bmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{bmatrix}$$
The determinant is
$$e^{2x}(cos^2 x + \sin^2 x) = e^{2x} > 0$$
Thus f' is invertible for all of $\mathbb{R}^2$

3. Recall from part 1 that we showed that the angle of (a,b) depended on $y + 2\pi n$. Thus we get
$$f(x, y{+}2\pi n) = (e^x \cos(y{+}2\pi n), e^x \sin(y{+}2\pi n)) = (e^x \cos(y), e^x \sin(y)) = f(x,y)$$
Thus f is not one-to-one.

5. Let $H := \{(x,y) \in \mathbb{R}^2 : y > 0\}$, and for $(x,y) \in H$ define
$$F(x,y) := \left(\frac{x^2+y^2-1}{x^2+2y+y^2+1}, \frac{-2x}{x^2+2y+y^2+1}\right)$$
Prove that F is a bijective mapping from H to $B(0,1)$, it's continuously differentiable on H, and it's inverse is also continuously differentiable.

Compute $|F(H)|$ we get
$$\sqrt{\left(\frac{x^2+y^2-1}{x^2+2y+y^2+1}\right)^2 + \left(\frac{-2x}{x^2+2y+y^2+1}\right)^2}$$

$$
\begin{aligned}
&= \frac{\sqrt{(x^2+y^2-1)^2+(-2x)^2}}{x^2+2y+y^2+1} \\
&= \frac{\sqrt{(x^2+y^2-1)^2+4x^2}}{x^2+2y+y^2+1} \\
&= \frac{\sqrt{x^4+y^4+2x^2y^2-2x^2-2y^2+1+4x^2}}{x^2+2y+y^2+1} \\
&= \frac{\sqrt{x^4+y^4+2x^2y^2-2y^2+1+2x^2}}{x^2+2y+y^2+1} \\
&= \frac{\sqrt{(x^2+y^2)^2-2y^2+1+2x^2}}{x^2+2y+y^2+1} \\
&< \frac{\sqrt{(x^2+y^2)^2+2y^2+1+2x^2}}{x^2+2y+y^2+1} \\
&= \frac{\sqrt{(x^2+y^2)^2+2(x^2+y^2)+1}}{x^2+2y+y^2+1} \\
&= \frac{x^2+y^2+1}{x^2+2y+y^2+1} \\
&< 1
\end{aligned}
$$

Thus we get $F(H) = B(0,1)$ To show $F(x,y)$ is surjective fix $(u,v) \in B(0,1)$. Write

$$
\begin{aligned}
u &= \frac{x^2+y^2-1}{x^2+2y+y^2+1} \\
v &= \frac{-2x}{x^2+2y+y^2+1}
\end{aligned}
$$

Rewriting we get

$$
u = \frac{x^2+y^2-1}{x^2+(y+1)^2} = \frac{x^2-1-2y-1+(y+1)^2}{x^2+(y+1)^2} = 1 - \frac{2y+2}{x^2+(y+1)^2}
$$

$$
v = \frac{-2x}{x^2+(y+1)^2}
$$

Then

$$
\begin{aligned}
(1-u)^2+v^2 &= \left(\frac{2y+2}{x^2+(y+1)^2}\right)^2 + \left(\frac{-2x}{x^2+(y+1)^2}\right)^2 \\
&= \frac{4((y^2+1)^2+x^2)}{(x^2+(y+1)^2)^2} \\
&= \frac{4}{x^2+(y+1)^2}
\end{aligned}
$$

Hence

$$
x^2+(y+1)^2 = \frac{4}{(1-u)^2+v^2}
$$

Recall that $v = \frac{-2x}{x^2+(y+1)^2}$. Thus we get

$$\begin{aligned} x &= v\frac{x^2+(y+1)^2}{-2} \\ &= v\frac{\frac{4}{(1-u)^2+v^2}}{-2} \\ &= \frac{-2v}{(1-u)^2+v^2} \end{aligned}$$

Substituting in x in the original equation we get.

$$\begin{aligned} \left(\frac{-2v}{(1-u)^2+v^2}\right)^2 + (y+1)^2 &= \frac{4}{(1-u)^2+v^2} \\ (y+1)^2 &= \frac{4((1-u)^2+v^2)}{((1-u)^2+v^2)^2} - \frac{4v^2}{((1-u)^2+v^2)^2} \\ (y+1)^2 &= \frac{4(1-u)^2}{((1-u)^2+v^2)^2} \\ y &= \frac{2(1-u)}{(1-u)^2+v^2} - 1 \\ y &= \frac{1-u^2+v^2}{(1-u)^2+v^2} \end{aligned}$$

Since $u^2+v^2<1$ we see that $1+v^2>u^2 \implies 1-u^2+v^2>0$ and thus $y>0$. Thus we found an explicit inverse of $F(x,y)=(u,v)$.
To show F is continuously differentiable, take the Jacobian (I used an online calculator for the latex in this step)

$$F(x,y) := \left(\frac{x^2+y^2-1}{x^2+2y+y^2+1}, \frac{-2x}{x^2+2y+y^2+1}\right)$$

$$\begin{aligned} F'(x,y) &= \begin{bmatrix} \frac{\partial}{\partial x}\frac{x^2+y^2-1}{x^2+2y+y^2+1} & \frac{\partial}{\partial y}\frac{x^2+y^2-1}{x^2+2y+y^2+1} \\ \frac{\partial}{\partial x}\frac{-2x}{x^2+2y+y^2+1} & \frac{\partial}{\partial y}\frac{-2x}{x^2+2y+y^2+1} \end{bmatrix} \\ &= \begin{bmatrix} \frac{4(y+1)x}{(x^2+(y+1)^2)^2} & \frac{2(y-x+1)(y+x+1)}{((y+1)^2+x^2)^2} \\ \frac{-2(y-x+1)(x+y+1)}{(x^2+(y+1)^2)^2} & \frac{4x(y+1)}{((y+1)^2+x^2)^2} \end{bmatrix} \end{aligned}$$

Taking the determinant

$$\begin{aligned} &\left(\frac{4(y+1)x}{(x^2+(y+1)^2)^2}\right)^2 - \left(\frac{2(y-x+1)(y+x+1)}{((y+1)^2+x^2)^2}\right)^2 \\ &= \frac{16x^2(y+1)^2 - 4(y-x+1)^2(y+x+1)^2}{((y+1)^2+x^2)^4} \end{aligned}$$

Which trivially, is strictly positive. Thus F' is invertible. I will generalize that $F^{-1}(u,v)$ is continuously differentiable.

5.1 Let $U, V \subset \mathbb{R}^n$ be open and suppose $f : U \to V$ is bijective, continuously differentiable, and everywhere invertible. Prove that $f^{-1} : V \to U$ is also continuously differentiable.

Fix $x_0 \in U$ and choose $y_0 = f(x_0) \in V$. Since $Df(x_0)$ is invertible and f is continuously differentiable, then there exist open neighborhoods

$$x_0 \in W \subset U \qquad y_0 \in Z \subset V$$

When we have a restrictive map $f_W : W \to Z$ and the bijection has a local C^1 inverse g. With the derivative formula being

$$Dg(y) = [D(f(g(y))]^{-1} \qquad y \in Z$$

Since f is a bijection, there is a well defined inverse $f^{-1} : V \to U$. On the neighborhood Z, g and f^{-1} are inverses of f. By injectivity of f, inverses are unique for each f and thus f^{-1} agrees with a C^1 map on every neighborhood for any arbitrary $y_0 \in V$. On each neighborhood Z we have

$$D(f^{-1}(y) = D(g(y)) = [D(f(g(y))]^{-1} = [D(f(f^{-1}(y))^{-1}]$$

Which gives us the global formula. Since f^{-1} is continuous and Df is continuous on U the matrix inversion map $J \to J^{-1}$ is continuous on $GL(n)$. Hence $f^{-1} \in C^1(V)$

6. Suppose $U \subset \mathbb{R}^2$ is an open set and $f : U \to \mathbb{R}$ is a C^1 function such that $\nabla f(x,y) \neq 0$ for all $(x,y) \in U$. Show that every level set is a C^1 smooth curve. That is for every $(x,y) \in U$, there exists a C^t function $\gamma : (-\delta, \delta) \to \mathbb{R}^2$ with $\gamma'(0) \neq 0$ such that $f(\gamma(t))$ is constant for all $t \in (-\delta, \delta)$.

Fix $(a,b) \in U$ and define

$$F(x,y) = f(x,y) - f(a,b)$$

Thus $F(a,b) = 0$ Since $\nabla f(a,b) \neq 0$ then at least one partial derivative is nonzero. WLOG assume

$$\frac{\partial F}{\partial y}(a,b) = f_y(a,b) \neq 0$$

By the implicit function theorem there exists open intervals I and J such that $a \in I$ and $b \in J$ then there exists a C_1 function $g : I \to J$ with

$$F(x, g(x)) = 0$$

Let $\gamma(t)$ be a parameterization of the level set near (a, b) where

$$\gamma(t) := (a + t, g(a + t))$$

Choose $\delta > 0$ such that $a + t \in I$ For some small $|t| < \delta$. Thus we get $(a - \delta, a + \delta) \in I$. Since $g \in C^1$ then $\gamma \in C^1$. Composing $F(\gamma(t))$ we get

$$F(\gamma(t)) = F(a + t, g(a + t)) = 0$$

$$f(\gamma(t)) = f(a, b)$$

which is constant. Also

$$\gamma'(t) = (1, g'(a + t)) \qquad \gamma'(0) \neq 0$$

Closing remarks

I think this was definitely harder than 150A because I had to go out and learn some new concepts and I eventually had to give in and look up a solution. My favorite part was just generalizing the problem and proving the stronger case. Like problem 6 from homework 3 or 5.1 (the problem I made) from the last homework. I took a long time to complete this as I was doing my